1. Write a program to initialize an integer array and print the sum and average of the array.

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr = {10,23,44,54,56};
        int Sum = 0;
        for(int num:arr) {
            Sum += num;
        }
        float Average = (float)Sum/arr.length;
        System.out.println("Average Of The Array: "+Average);
        System.out.print("Sum Of The Array: "+Sum);
    }
}
```

```
Average Of The Array: 37.4
Sum Of The Array: 187
```

2. Write a program to initialize an integer array and find the maximum and minimum value of the array.

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr = {11,45,65,10,18,32,22,85,17,23};
        int Max = 0;
        int Min = Integer.MAX_VALUE;
        for(int num:arr) {
            if(num>Max) {
                Max = num;
            }
            else if(num<Min) {
                Min = num;
            }
        }
        System.out.println("Maximum of the array :"+Max);
        System.out.println("Minimum of the array :"+Min);
    }
}
```

```
Maximum of the array :85
Minimum of the array :10
```

3.Write a program to initialize an integer array with values and check if a given number is present in the array or not. If the number is not found, it will print -1 else it will print the index value of the given number in the array. Example 1) If the Array elements are {1,4,34,56,7} and the search element is 90

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr =  {1,4,34,56,7};
        int target =  56;
        int l = 0;
        int r = arr.length-1;
        boolean flag = true;
        while(l<r) {
            int mid = (l+r)/2;
            if(arr[mid]==target) {
                System.out.print("Search Index : "+mid);
                flag = false;
                break;
            }
            else if(arr[mid]<target) {
                l = mid+1;
            }
            else {
                r = mid-1;
            }
        }
        if(flag) {
            System.out.print("Search Index: "+-1);
        }
    }
}
```

```
Search Index : 3
```

4.Initialize an integer array with ascii values and print the corresponding character values in a single row.

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr =  {65,66,67,68,69,70,71,72,73,74,75};
        System.out.print("Uppercase Letters : ");
        for(int num:arr) {
            System.out.print((char)('A'+(num-'A'))+" ");
        }
        System.out.println();
        int[] Arr =  {97,98,99,100,101,102,103,104,105,106,107};
        System.out.print("Lowercase Letters : ");
        for(int num:Arr) {
            System.out.print((char)('a'+(num-'a'))+" ");
        }
    }
}
```

```
Uppercase Letters : A B C D E F G H I J K
Lowercase Letters : a b c d e f g h i j k
```

5.Write a program to initialize an array and print them in a sorted order.

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr = {10,20,45,1,30,4,3,2,0,5};
        for(int i=0;i<arr.length;i++) {
            for(int j=1;j<arr.length-i;j++) {
                if(arr[j]<arr[j-1]) {
                    int temp = arr[j];
                    arr[j] = arr[j-1];
                    arr[j-1] = temp;
                }
            }
        }
        System.out.print(Arrays.toString(arr));
    }
}
```

```
[0, 1, 2, 3, 4, 5, 10, 20, 30, 45]
```

6.Write a program to remove the duplicate elements in an array and print the same. Example)
I/P:{12,34,12,45,67,89} O/P:{12,34,45,67,89}

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr = {12,34,12,45,67,89};
        HashSet<Integer> set = new LinkedHashSet<>();
        for(int i=0;i<arr.length;i++){
            set.add(arr[i]);
        }
        System.out.print(set);
    }
}
```

```
[12, 34, 45, 67, 89]
```

7.Write a program to find the largest 2 numbers and the smallest 2 numbers in the given array.

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int[] arr = {3,1,2,4,5,6,7,8,9,10};
        int largest = Integer.MIN_VALUE;
        int secondLargest = Integer.MIN_VALUE;
        int smallest = Integer.MAX_VALUE;
        int secondSmallest = Integer.MAX_VALUE;
        for(int num : arr){
            if(num > largest){
                secondLargest = largest;
                largest = num;
            }
            else if(num > secondLargest && num != largest){
                secondLargest = num;
            }
            if(num < smallest){
                secondSmallest = smallest;
                smallest = num;
            }
            else if(num < secondSmallest && num != smallest){
                secondSmallest = num;
            }
        }
        System.out.println("Largest Element : " + largest);
        System.out.println("Second Largest Element : " + secondLargest);
        System.out.println("Smallest Element : " + smallest);
        System.out.println("Second Smallest Element : " + secondSmallest);
    }
}
```

```
Largest Element : 10
Second Largest Element : 9
Smallest Element : 1
Second Smallest Element : 2
```

8.Write a program to reverse the elements of a given 2*2 array. Four integer numbers needs to be passed as Command Line arguments. Example1) C:\>java Sample 1 2 3 O/P: Please enter 4 integer numbers Example2) C:\>java Sample 1 2 3 4 O/P: The given array is : 1 2 3 4 The reverse of the array is : 4 3 2 1

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int row = 2;
        int col = 2;
        int index = (row*col)-1;
        for(int i=row-1;i>=0;i--){
            for(int j=col-1;j>=0;j--){
                System.out.print(args[index--]+" ");
            }
            System.out.println();
        }
    }
}
```

```
4 3
2 1
```

9.Write a program to find the biggest number in a 3*3 array. The program is supposed to receive 9 integer numbers as command line arguments. Example1: C:\>java Sample 1 2 3 O/P: Please enter 9 integer numbers Example2: C:\>java Sample 1 23 45 55 121 222 56 77 89 O/P: The given array is : 1 23 45 55 121 222 56 77 89 The biggest number in the given array is 222

```java
package main;
import java.util.*;
public class Main {
    public static void main(String[] args) {
        int row = 3;
        int col = 3;
        int max = Integer.parseInt(args[0]);
        int index = 0;
        for(int i = 0; i < row; i++){
            for(int j = 0; j < col; j++){
                int num = Integer.parseInt(args[index++]);
                if(num > max){
                    max = num;
                }
            }
        }
        System.out.println("Largest number: " + max);
    }
}
```

```
Largest number: 222
```